THE GEOMETRIC FIBRE FUNCTOR ON THE SMALL ETALE SITE IS THE STALK AT A POINT OF THE SITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own native preamble excerpt used for this source: the Xy-pic setup at lines 14-18 and the theorem declarations with the mathematical macros at lines 154-217 of sources/stacks-etale-cohomology/preamble.tex. The quoted class line and the xr-hyper externaldocument block are not included because the assembler supplies a plain amsart wrapper; no macro was invented or redefined.

Editorial note. The same-chapter reference is located at its \label line inside the pinned chapter file; the three cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Point of a site and its associated stalk functor. Sites, Definition sites-definition-point (tag 00Y5) and Sites, Equation sites-equation-stalk (tag 04EH), named in the statement at lines 3383-3385; the criterion for a functor to be a point is Sites, Proposition sites-proposition-point-limits (tag 00YC), invoked at line 3406. Supply the definition and the criterion that has to be verified; the verification is the counted content.
- Every scheme etale over an algebraically closed field splits as a disjoint union of copies of the field, and lifting of geometric points along etale coverings. Same chapter, Lemma labelled lemma-geometric-lift-to-cover (source line 3305, Stacks tag 03PR), used at proof lines 3391-3392 and again at 3403-3404. Supplies that U over the geometric point is a disjoint union of copies of the point, and that a geometric point of an object lifts to some member of any etale covering; the functoriality and covering computations are retained in the counted span.

Source reference. Original definition 00Y5: sources/stacks-chow/tags.txt, source lines 1243–1243. Cross-chapter pointer named in the counted statement at line 3383 for the notion of a point of a site; the verification that u defines such a point is retained verbatim in the counted span.

Source reference. Original equation 04EH: sources/stacks-chow/tags.txt, source lines 5763–5763. Cross-chapter pointer named twice in the counted statement (lines 3385 and 3411) for the displayed formula defining the stalk functor; the comparison of the two colimits is retained in the counted span.

Source reference. Original proposition 00YC: sources/stacks-chow/tags.txt, source lines 1250–1250. Cross-chapter standard prerequisite invoked once at proof line

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3406 as the criterion for a functor to define a point of a site; the three conditions it requires are verified inside the counted span.

Source reference. Original definition 03PO: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 3201–3201. Same-chapter pointer invoked at proof line 3398 for the description of a geometric point as a morphism from the spectrum of an algebraically closed field; the identification of the underlying set of the geometric fibre with that collection is retained in the counted span.

Source reference. Original lemma 03PR: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 3305–3305. Same-chapter pointer invoked at proof lines 3391-3392 and 3403-3404 for the shape of the geometric fibre and for surjectivity on coverings; both uses are inputs whose consequences are drawn inside the counted span.

Lemma 0.1. *Let S be a scheme. Let $\overline{s}$ be a geometric point of S. Consider the functor*

$$u : S_{étale} \longrightarrow Sets,$$
$$U \longmapsto |U_{\overline{s}}| = \{\overline{u} \text{ such that } (U, \overline{u}) \text{ is an étale neighbourhood of } \overline{s}\}.$$

Here $|U_{\overline{s}}|$ denotes the underlying set of the geometric fibre. Then u defines a point p of the site $S_{étale}$ (Sites, Definition 00Y5) and its associated stalk functor $\mathcal{F} \mapsto \mathcal{F}_p$ (Sites, Equation 04EH) is the functor $\mathcal{F} \mapsto \mathcal{F}_{\overline{s}}$ defined above.

Proof. In the proof of Lemma 03PR we have seen that the scheme $U_{\overline{s}}$ is a disjoint union of schemes isomorphic to $\overline{s}$. Thus we can also think of $|U_{\overline{s}}|$ as the set of geometric points of U lying over $\overline{s}$, i.e., as the collection of morphisms $\overline{u} : \overline{s} \to U$ fitting into the diagram of Definition 03PO. From this it follows that $u(S)$ is a singleton, and that $u(U \times_V W) = u(U) \times_{u(V)} u(W)$ whenever $U \to V$ and $W \to V$ are morphisms in $S_{étale}$. And, given a covering $\{U_i \to U\}_{i \in I}$ in $S_{étale}$ we see that $\coprod u(U_i) \to u(U)$ is surjective by Lemma 03PR. Hence Sites, Proposition 00YC applies, so p is a point of the site $S_{étale}$. Finally, our functor $\mathcal{F} \mapsto \mathcal{F}_{\overline{s}}$ is given by exactly the same colimit as the functor $\mathcal{F} \mapsto \mathcal{F}_p$ associated to p in Sites, Equation 04EH which proves the final assertion. □